
Machine Learning and QUBO-Based Portfolio Optimization: A Comparative Study on NIFTY50 Equities

Abstract

This report compiles five portfolio optimization analyses, progressing from the classical Markowitz mean-variance framework to convex-optimization solutions, machine-learning-enhanced return estimation, and finally quantum-inspired Quadratic Unconstrained Binary Optimization (QUBO) formulations solved via simulated annealing. Each method is briefly explained along with its main results. The report concludes with a benchmark section in which all methods are compared using the same unseen test data to evaluate their real-world performance side by side. The results indicate that the covariance-aware, cardinality-constrained QUBO portfolio (10 equally weighted assets) achieved the highest realized out-of-sample Sharpe ratio among the methods considered in both asset universes tested (0.878 vs. 0.68 for ML Markowitz on the Full Universe; 1.141 vs. 0.79 on the Extended Universe). At the same time, the resulting portfolios were more concentrated and exhibited greater sensitivity to the risk-aversion parameter γ than the classical and ML-enhanced continuous-weight Markowitz approaches.

Executive Summary

The analysis in this report is organized around a progression from continuous portfolio optimization to discrete, cardinality-constrained formulations. Starting with the classical Markowitz framework, the study first establishes a convex-optimization baseline, then examines whether machine-learning-based return estimates can improve portfolio construction. The analysis subsequently reformulates asset selection as a QUBO problem and extends this formulation to incorporate covariance risk. All approaches are ultimately evaluated under a common out-of-sample framework, allowing their realized performance to be compared on a consistent basis.

Main contributions

- Established classical Markowitz optimization as a continuous-weight benchmark for NIFTY 50 equities.
- Used machine learning to improve the estimation and cross-sectional ranking of expected returns.
- Reformulated fixed-cardinality asset selection as a Quadratic Unconstrained Binary Optimization (QUBO) problem.
- Extended the QUBO formulation to incorporate covariance-based portfolio risk.
- Compared continuous-weight and QUBO-based approaches using a common out-of-sample evaluation protocol.

Key findings

The covariance-aware QUBO formulation produced the strongest realized risk-adjusted performance among the methods considered in both asset universes. However, this advantage was accompanied by greater portfolio concentration and stronger sensitivity to the risk-aversion parameter γ . The results therefore suggest that the QUBO formulation can be effective when discrete asset-selection constraints are important, while also highlighting the trade-offs introduced by restricting the portfolio to a fixed number of holdings.

Scope and limitations

The empirical comparison is based on a single rebalance date and one out-of-sample holding window. The QUBO experiments use simulated annealing rather than quantum hardware, and the analysis does not incorporate transaction costs or periodic portfolio rebalancing. These limitations are important when interpreting the reported performance and motivate the extensions discussed in the final sections of the report.

Methodology

The analysis follows a common pipeline that allows the different portfolio construction approaches to be evaluated on a consistent basis. As illustrated in Figure 1, historical price data are first transformed into predictive features and used to estimate future returns through the machine-learning component. These return estimates then serve as inputs to two optimization branches: continuous-weight Markowitz optimization and discrete QUBO-based portfolio selection.

The QUBO branch is solved using simulated annealing and is subsequently extended to incorporate covariance-based portfolio risk. The resulting portfolios, together with the classical and ML-enhanced Markowitz portfolios, are evaluated using the same out-of-sample data and performance measures. This common evaluation framework forms the basis for the head-to-head comparison presented later in the report.

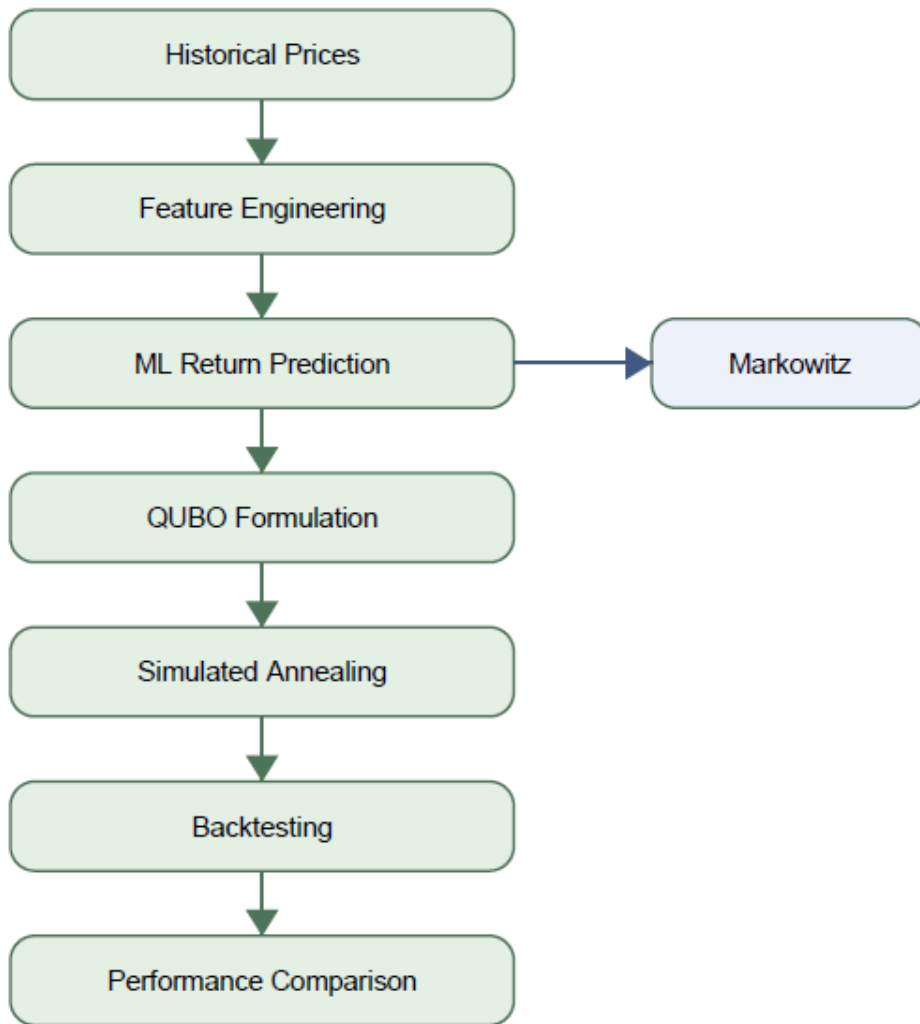


Figure 1: Overview of the portfolio optimization methodology. Historical price data are transformed into predictive features and used for return estimation. The resulting estimates feed the Markowitz and QUBO optimization branches, which are evaluated under a common out-of-sample backtesting framework.

The detailed analysis proceeds in stages. The Markowitz framework establishes the theo-

retical foundation and the continuous optimization baseline. Machine learning is then introduced to improve expected-return estimation. The analysis subsequently considers cardinality-constrained QUBO selection before extending the formulation to incorporate portfolio covariance and risk. Finally, the benchmark section compares the resulting approaches using their realized out-of-sample performance.

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1 Introduction

Constructing an optimal investment portfolio requires balancing expected return against risk. Harry Markowitz’s mean-variance framework, introduced in 1952, remains the conceptual starting point for almost all modern portfolio construction. This report compiles a five-part investigation into the Markowitz problem and its modern solution methods, applied throughout to a universe of NIFTY 50 constituent stocks:

- Part 1: Theory.** A conceptual treatment of the Markowitz problem and its two equivalent formulations (risk-constrained and risk-adjusted).
- Part 2: Classical solver.** A convex-optimization implementation using historical mean returns, solved via the Clarabel conic solver [4].
- Part 3: Machine learning.** Replacing historical mean returns with ML-predicted returns (pooled cross-sectional regression) as inputs to the same mean-variance pipeline, evaluated out-of-sample.
- Part 4: QUBO (cardinality only).** Selecting a fixed subset of exactly $K = 10$ assets to maximize predicted return as a Quadratic Unconstrained Binary Optimization problem, solved exactly via brute force and approximately via simulated annealing [6].
- Part 5: QUBO with risk.** Extending the QUBO formulation to a full mean-variance objective (return $- \gamma \cdot \text{risk} + \lambda \cdot \text{cardinality penalty}$), evaluated out-of-sample across a range of risk-aversion values γ , plus a discretized-weight stress test that allows the QUBO to approximate continuous portfolio weights.

Data. The analysis uses daily adjusted closing prices for constituents of the NIFTY 50 index obtained from the Kaggle dataset *NIFTY50 Stock Market Data* [2]¹. To accommodate differences in listing histories and data availability, the stocks are grouped into two asset universes:

- **Full Universe:** 50 stocks with approximately 10 years of overlapping price history.
- **Extended Universe:** 38 stocks with approximately 17 years of available price history, providing a longer time horizon at the expense of a smaller cross-section of assets.

2 The Markowitz Framework

2.1 Core Idea

Markowitz’s central insight is that portfolio risk depends not only on individual asset volatility but also on how returns co-move, formalizing the benefit of diversification [1]. Let $\mu \in \mathbb{R}^n$ be the vector of expected returns, $\Sigma \in \mathbb{R}^{n \times n}$ the covariance matrix, and $w \in \mathbb{R}^n$ the portfolio weight vector. Then

$$\mathbb{E}[R] = \mu^\top w, \tag{1}$$

$$\text{Var}(R) = w^\top \Sigma w. \tag{2}$$

¹<https://www.kaggle.com/datasets/rohanrao/nifty50-stock-market-data>

2.2 Two Equivalent Formulations

Risk-constrained form (maximize return subject to a risk budget):

$$\max_w \mu^\top w \quad (3)$$

$$\text{s.t. } w^\top \Sigma w \leq \sigma_{\text{target}}^2, \quad \mathbf{1}^\top w = 1. \quad (4)$$

Risk-adjusted form (combine return and risk in one objective):

$$\max_w \mu^\top w - \gamma w^\top \Sigma w \quad (5)$$

$$\text{s.t. } \mathbf{1}^\top w = 1. \quad (6)$$

These are mathematically equivalent: the Lagrange multiplier on the variance constraint in the first formulation plays the role of the risk-aversion parameter γ in the second. Varying σ_{target} or γ traces out the same efficient frontier, the choice between them is interpretational, not fundamental. This same risk-adjusted form reappears, in binary-variable disguise, in the QUBO formulations of Sections 5 and 6.

2.3 Limitations and Modern Extensions

The classical Markowitz model relies heavily on accurate estimates of expected returns and covariances, which can be difficult to obtain in practice. It also assumes that positive and negative deviations from the expected return are treated equally and considers only a single investment period. These limitations motivate the alternative approaches explored in this report, including constrained convex optimization (Section 3), machine learning based return prediction (Section 4), and QUBO-based formulations that can naturally incorporate discrete constraints such as limiting the number of selected assets (Sections 5–6).

3 Classical Convex Optimization

3.1 Setup

The risk-constrained Markowitz problem,

$$\max_w \mu^\top w \quad \text{s.t. } w^\top \Sigma w \leq \sigma^2, \quad \mathbf{1}^\top w = 1, \quad w \geq 0, \quad (7)$$

is convex and was solved using **Clarabel** [4], an interior-point conic solver, implemented both via CVXPY [3] and via Clarabel’s raw cone interface (risk constraint expressed as a second-order cone via Cholesky factorization of Σ). Both approaches produced identical solutions. The Global Minimum Variance (GMV) risk floors were $\approx 13.4\%$ (Full Universe) and $\approx 17.0\%$ (Extended Universe).

3.2 Results

Table 1: Full Universe (50 stocks): expected return at target risk levels

Target Risk	Exp. Return	# Stocks	Key Holdings
15%	22.04%	12	HINDUNILVR, SHREECEM
20%	29.82%	6	HINDUNILVR, BAJAJFINSV
25%	33.56%	5	BAJAJFINSV, SHREECEM

Table 2: Extended History (38 stocks): expected return at target risk levels

Target Risk	Exp. Return	# Stocks	Key Holdings
17%	16.70%	21	HINDUNILVR, BRITANNIA
20%	28.13%	12	SHREECEM, ASIANPAINT
25%	35.08%	7	SHREECEM, BAJFINANCE

Note: The reported returns are the optimizer’s estimated expected returns based on historical data at the rebalancing date. They are shown for illustration and are not directly comparable to the out-of-sample performance reported later.

Higher risk targets concentrate the portfolio into fewer, higher-return names. A 10% maximum-weight cap at the 20% risk target increased diversification from 6 to 12 stocks while reducing expected return from 29.82% to 25.86%. Robustness checks (solver tolerance, scaling, regularization, input noise) confirmed that optimal portfolios are numerically stable and that default Clarabel settings suffice.

Figure 2 compares the efficient frontiers of both asset universes, and Figure 3 shows the Full Universe Monte Carlo simulation overlaid with the convex-optimization frontier, confirming that the solver finds solutions on or above the Monte Carlo envelope. Figure 4 illustrates how imposing a maximum weight cap trades off expected return against diversification.

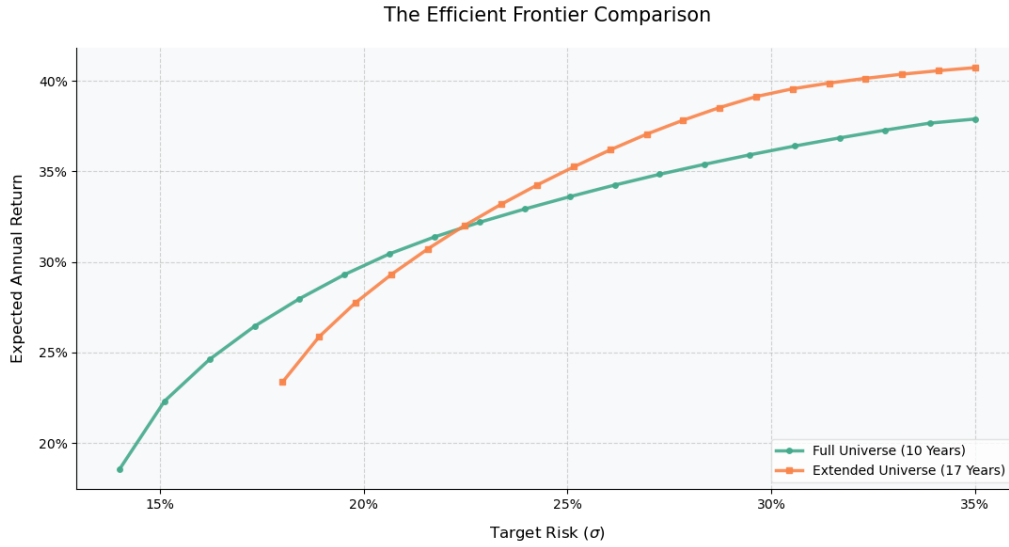


Figure 2: Efficient frontier comparison between the Full Universe (50 stocks, 10 years) and Extended Universe (38 stocks, 17 years). The Extended Universe achieves higher expected returns at equivalent risk levels due to its longer estimation window capturing a different return regime.

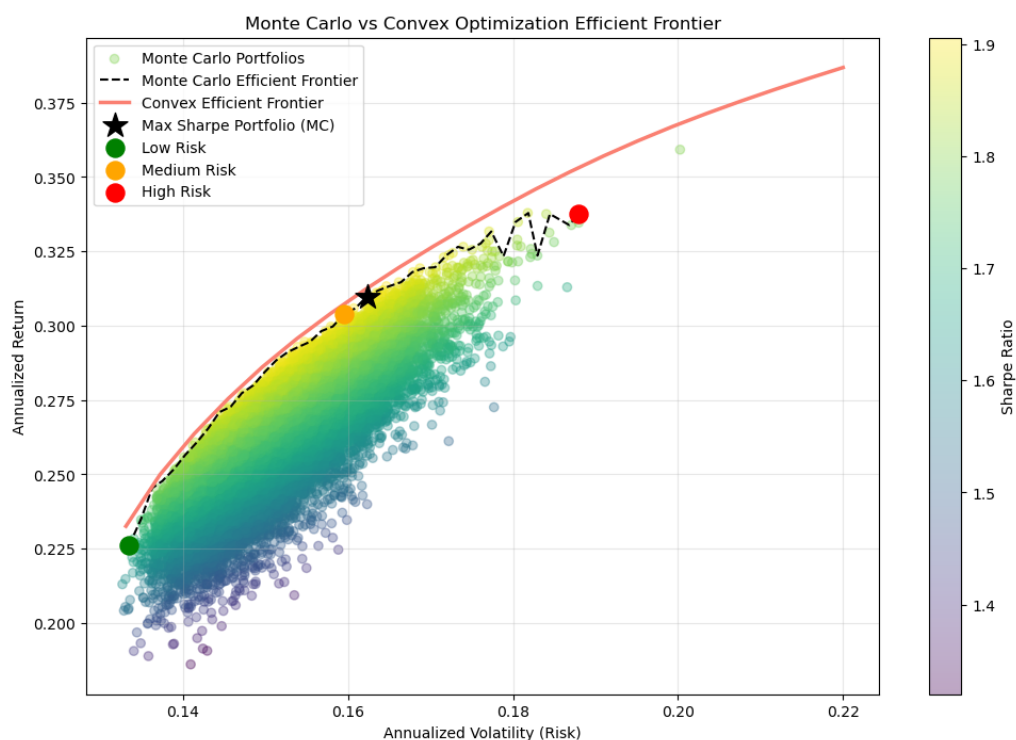


Figure 3: Monte Carlo simulation (5,000 random portfolios, Full Universe) vs. the convex-optimization efficient frontier. The convex frontier lies above and to the left of the Monte Carlo envelope, confirming that the Clarabel solver identifies genuinely optimal portfolios. The star marks the maximum Sharpe ratio portfolio from Monte Carlo; colored dots indicate low-, medium-, and high-risk reference portfolios.

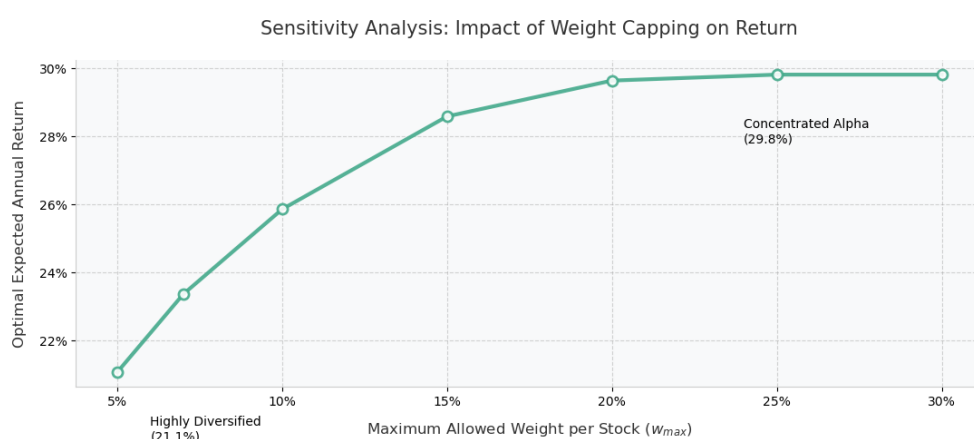


Figure 4: Sensitivity of optimal expected annual return to the maximum allowed weight per stock (w_{max}) at a 20% risk target, Full Universe. Tighter caps force greater diversification at the cost of lower expected return; the curve flattens beyond $w_{max} \approx 20\%$, indicating diminishing benefit from further concentration.

4 Machine-Learning-Enhanced Portfolio Construction

4.1 Setup

A pooled cross-sectional linear regression predicts 15-day forward cumulative returns $R_{t+1:t+15} = \sum_{i=1}^{15} r_{t+i}$, using lagged returns (1, 5, 10, 15 days), rolling momentum (5, 10, 20 days), and rolling volatility (10, 20 days), all constructed from information available strictly before the forecast origin. Training uses data before 2019-01-01; later dates are held out for testing, with a 15-day buffer to prevent target–test overlap. Out-of-sample MSE was **0.01137** (Full) and **0.01247** (Extended), predictive accuracy is moderate, but the resulting cross-sectional ranking is informative enough to be useful as an input to mean-variance optimization.

ML-predicted returns are converted to daily expected returns (dividing the 15-day forecast by 15), annualized, and combined with an annualized sample covariance estimated on training data only. Portfolios are constructed once, at the first test-date rebalance (2019-01-01), and held fixed thereafter.

4.2 Out-of-Sample Performance

Table 3: Out-of-Sample Performance: Full Universe

Portfolio (Risk Target)	Return (%)	Volatility (%)	Sharpe
ML Markowitz (0.12)	13.15	19.36	0.68
Classic Markowitz (0.12)	13.49	20.02	0.67
Equal Weight	15.06	24.33	0.62
ML Markowitz (0.15)	11.62	22.20	0.52
Classic Markowitz (0.15)	11.41	26.35	0.43
ML Markowitz (0.20)	11.99	24.75	0.48
Classic Markowitz (0.20)	5.75	37.73	0.15

Table 4: Out-of-Sample Performance: Extended Universe

Portfolio (Risk Target)	Return (%)	Volatility (%)	Sharpe
ML Markowitz (0.17)	17.03	21.56	0.79
Classic Markowitz (0.17)	16.24	21.94	0.74
Equal Weight	14.39	25.22	0.57
ML Markowitz (0.20)	14.55	23.80	0.61
Classic Markowitz (0.20)	14.99	26.20	0.57
ML Markowitz (0.25)	14.42	27.20	0.53
Classic Markowitz (0.25)	15.92	31.74	0.50

Figure 5 shows the out-of-sample cumulative return paths for the Full Universe at the best-performing risk target (0.12). All three portfolios experienced significant drawdowns during the COVID-19 market shock in early 2020 before recovering; the ML and Classic Markowitz portfolios recovered more smoothly than the equal-weight benchmark due to their lower volatility allocations.

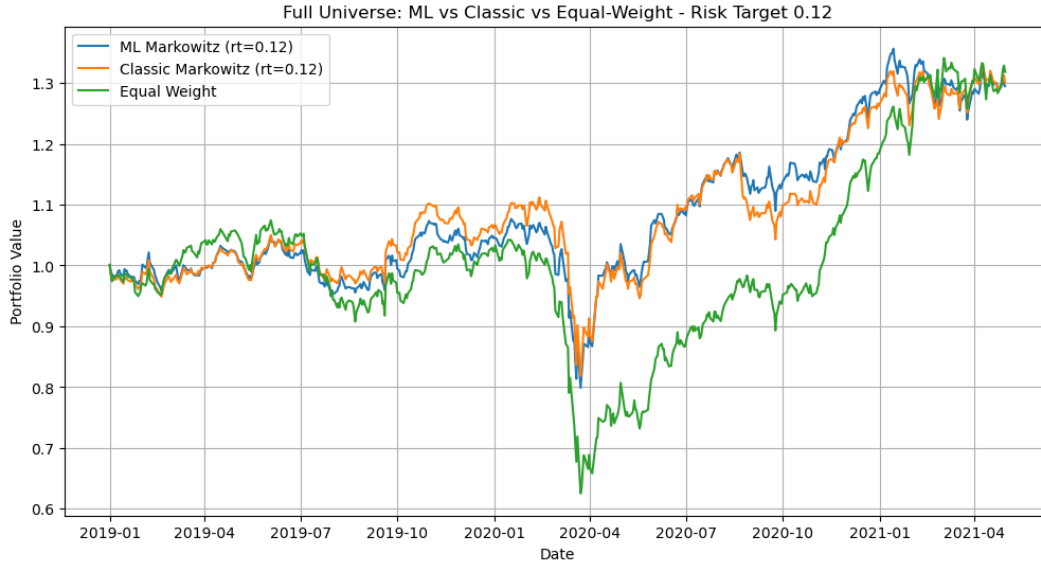


Figure 5: Out-of-sample cumulative portfolio value (Full Universe, risk target $\sigma = 0.12$) for ML Markowitz, Classic Markowitz, and Equal Weight from January 2019 to mid-2021. The sharp drawdown in March 2020 reflects the COVID-19 market shock.

Figure 6 shows the corresponding cumulative return paths for the Extended Universe at its best-performing risk target (0.17). The ML and Classic Markowitz portfolios track closely here, both outperforming the equal-weight benchmark by end of period, consistent with the Sharpe ratios in Table 4.

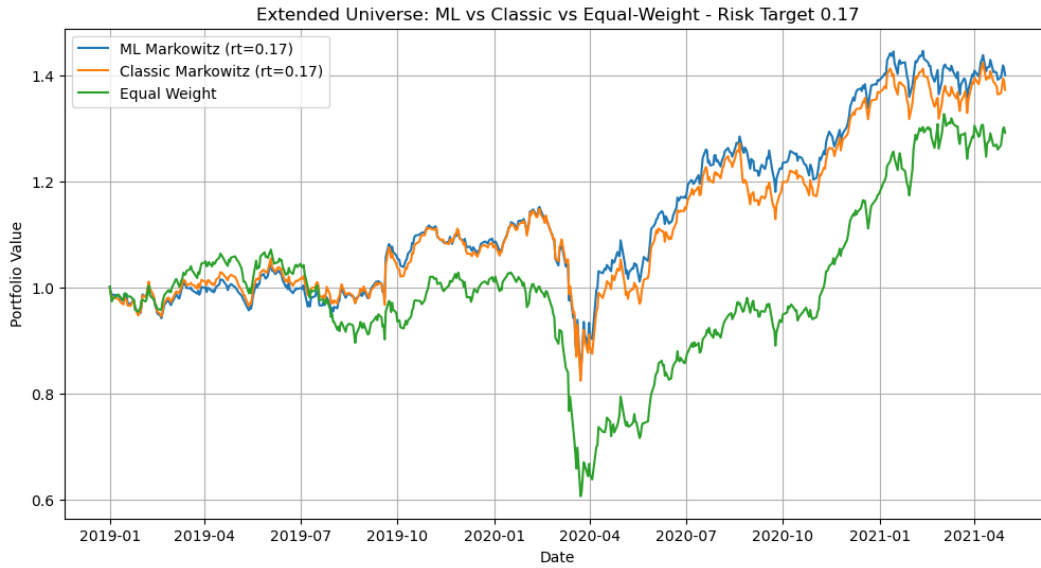


Figure 6: Out-of-sample cumulative portfolio value (Extended Universe, risk target $\sigma = 0.17$) for ML Markowitz, Classic Markowitz, and Equal Weight from January 2019 to mid-2021. ML and Classic Markowitz perform similarly at this risk level, both recovering more completely from the COVID-19 drawdown than the equal-weight benchmark.

The two cumulative-return plots reinforce the table-level result: the ML enhancement is most useful as a return-ranking signal within a diversified optimizer, rather than as a source of uniformly higher raw returns. The similar trajectories of the ML and classical portfolios also show that covariance structure and the risk constraint remain major determinants of realized

behavior.

At low risk targets, ML Markowitz modestly outperforms Classic Markowitz on Sharpe ratio in both universes; as the risk target relaxes, both optimizer-based portfolios become increasingly concentrated (effective number of holdings N_{eff} falls from ~ 14 -15 down to ~ 4 at the highest risk targets), and ML's edge narrows or reverses depending on universe. Concentration is similar across ML-based and classical portfolios, indicating the gain comes from better cross-sectional *ranking* of expected returns rather than from materially different diversification behavior.

4.3 Where ML Helps vs. Hurts

One-day-ahead return forecasting was found to be largely uninformative (extremely weak predictability, noisy inputs that produced unstable allocations). Shifting to a 15-day forward horizon reduced noise and produced more stable, economically meaningful expected-return estimates, yielding competitive and at low risk targets improved risk-adjusted performance relative to the classical benchmark. The gains, however, are moderate rather than transformative: diversification remains the dominant driver of realized performance, and ML primarily adds value by improving the cross-sectional ranking used to allocate that diversification.

5 QUBO-Based Cardinality-Constrained Selection

5.1 Motivation and Formulation

Real portfolios are often subject to a fixed holding count (e.g. “select exactly 10 names”), a combinatorial constraint that does not fit naturally into the continuous-weight Markowitz program. This is instead formulated as a **Quadratic Unconstrained Binary Optimization (QUBO)** problem [8,9]: binary decision variables $x_i \in \{0, 1\}$ indicate whether asset i is selected, and the cardinality constraint is folded into the objective as a quadratic penalty,

$$\min_x x^\top Q x, \quad Q_{ii} = -r_i - 19\lambda, \quad Q_{ij} = 2\lambda \quad (i \neq j), \quad (8)$$

where r_i are ML-predicted returns (from Section 4) and λ penalizes deviation from $K = 10$ selected assets. This is a *return-only* objective, risk is not yet incorporated (that extension is the subject of Section 6).

5.2 Exact Solution via Brute Force

On a reduced 15-asset subproblem ($2^{15} = 32,768$ candidate portfolios, exhaustively enumerable), brute force confirms the QUBO recovers the exact top-10 portfolio under the predicted-return objective in both universes:

Table 5: Brute-force QUBO optimum (reduced problem)

Universe	Energy	# Assets Selected
Full (reduced)	−108.27	10
Extended (reduced)	−155.02	10

Figure 7 shows how the penalty parameter λ governs this behavior across three diagnostics: the number of selected assets, the minimum QUBO energy, and the portfolio return each as a function of λ , confirmed via brute force on the reduced problem. Figure 8 shows the energy landscape as a function of portfolio size k , with the global minimum correctly occurring at $k = 10$.

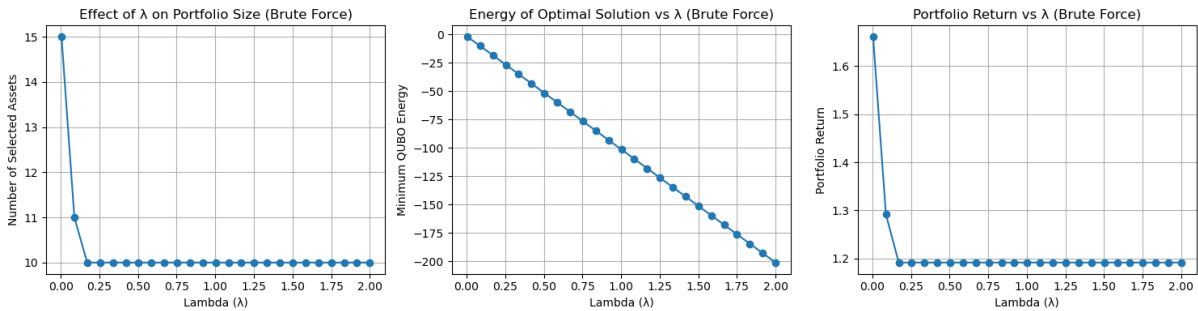


Figure 7: Effect of the penalty parameter λ on QUBO solutions (brute force, reduced 15-asset problem). *Left:* Number of selected assets as a function of λ , the constraint is enforced exactly at $k = 10$ once $\lambda \gtrsim 0.25$. *Center:* Minimum QUBO energy decreases roughly linearly in λ due to a constant penalty-induced shift. *Right:* Portfolio return plateaus once λ is large enough to dominate the objective, illustrating the risk of over-penalizing.

As the penalty λ increases from near-zero, the number of selected assets falls from “many” (weak penalty, most predicted returns positive) and stabilizes exactly at 10; minimum energy decreases roughly linearly in λ (a constant penalty-induced shift); and the energy landscape, scanned over portfolio size k , has its global minimum at $k = 10$, confirming the constraint is correctly enforced.

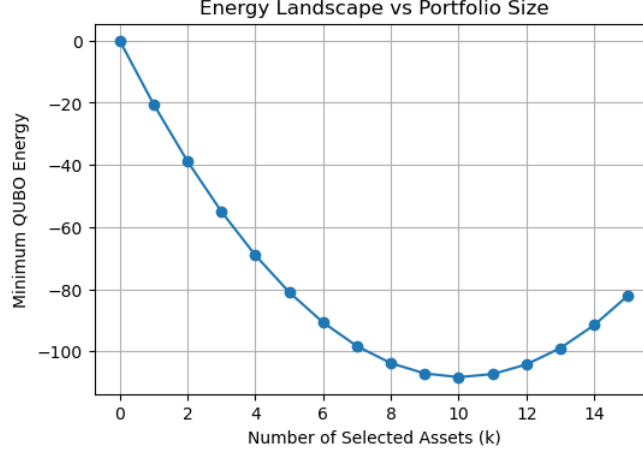


Figure 8: QUBO energy landscape as a function of portfolio size k (brute force, reduced 15-asset problem). The global minimum occurs at $k = 10$, confirming that the cardinality constraint is correctly encoded. The energy rises steeply for small k (return signal dominates) and more gradually for large k (penalty dominates).

5.3 Simulated Annealing on the Full Problem

Brute force is infeasible at full scale ($2^{50} \approx 10^{15}$), so simulated annealing (1275 QUBO terms, Full; 741 terms, Extended) is used instead:

Table 6: Simulated annealing vs. greedy top-10 (predicted-return objective, not annualized/backtested)

Universe	Top-10 Greedy	Simulated Annealing	% of Optimal
Full	1.2766	1.2049	~94%
Extended	1.8610	1.7866	~96%

Note: The reported “Portfolio Return” values are the returns estimated by the QUBO model at the rebalancing date. They are not realized out-of-sample returns and should not be compared directly with the Sharpe-ratio results reported in Sections 4, 6, and 7.

In both universes, simulated annealing selects 6 of the same 10 assets as the greedy top-10 solution while satisfying the cardinality constraint exactly, reaching 94–96% of the optimal predicted-return objective, a reasonable approximation given the combinatorial search space.

5.4 Effect of the Penalty Parameter λ

Table 7: Effect of λ on the QUBO solution (illustrative sweep)

λ	Energy	# Assets	Return
0.1	-11.28	11	1.378
0.5	-51.21	10	1.215
1.0	-100.00	10	1.089
5.0	-500.00	10	1.065
100.0	-10000.00	10	1.066

Small λ leaves the constraint too weak (more than 10 assets selected); once λ is large enough to dominate, the constraint is enforced exactly and the return contribution becomes secondary, plateauing as λ grows further. Proper tuning of λ is therefore critical: too small under-constrains the problem, too large can crowd out the return signal entirely. This observation directly motivated reintroducing risk explicitly into the objective (Section 6) rather than relying on the penalty term alone to shape the solution.

6 QUBO with Covariance

6.1 Motivation

Section 5 showed that a pure return-maximizing QUBO needs careful tuning of the penalty λ and ignores risk entirely. This section extends the formulation to a genuine **mean-variance QUBO**: a cardinality-constrained binary selection problem whose objective also penalizes portfolio variance, evaluated using the same out-of-sample backtest protocol as the ML/Classic Markowitz comparison in Section 4.

6.2 Formulation

With binary variables $x_i \in \{0, 1\}$, the objective is

$$\min_x \left(-\sum_i \mu_i x_i + \gamma \sum_{i,j} \Sigma_{ij} x_i x_j + \lambda \left(\sum_i x_i - K \right)^2 \right). \quad (9)$$

Expanding the cardinality penalty and the covariance term (using $x_i^2 = x_i$ for binary variables) and collecting coefficients of the upper-triangular QUBO matrix Q gives, for $K = 10$:

$$Q_{ii} = -\mu_i + \gamma \Sigma_{ii} - 19\lambda, \quad (10)$$

$$Q_{ij} = 2\gamma \Sigma_{ij} + 2\lambda \quad (i < j). \quad (11)$$

Here μ_i are the ML-predicted returns from Section 4 and Σ is the same annualized sample covariance matrix used throughout. λ is set to ten times the average magnitude of predicted returns, separately per universe, to keep the cardinality penalty commensurate with the return scale.

6.3 Solving and Backtesting

The QUBO is solved with the **neal** simulated-annealing sampler [7] (1000 reads) across a sweep of risk-aversion values $\gamma \in \{0.05, 0.5, 5, 15, 30, 60, 100\}$. For each γ , the selected 10 assets are equally weighted and the resulting portfolio is evaluated out-of-sample using the identical backtest window and methodology as Section 4 (so that the Sharpe ratios below are directly comparable to Tables 3–4).

Table 8: QUBO Mean-Variance Portfolio: Out-of-Sample Performance, Full Universe

γ	Annualized Return	Annualized Volatility	Sharpe
0.05	9.18%	23.23%	0.395
0.5	5.90%	21.88%	0.270
5	14.30%	21.89%	0.653
15	16.86%	21.30%	0.792
30	17.51%	19.95%	0.878
60	8.85%	20.34%	0.435
100	8.01%	19.78%	0.405

Figures 9 and 10 show cumulative return paths for all γ values across both universes. The non-monotonic ordering of curves as γ increases is clearly visible: intermediate values of γ do not always select better asset combinations than more extreme settings, because the 10-asset selection is discrete and equally weighted.

Table 9: QUBO Mean-Variance Portfolio: Out-of-Sample Performance, Extended Universe

γ	Annualized Return	Annualized Volatility	Sharpe
0.05	11.77%	25.50%	0.461
0.5	18.71%	21.71%	0.862
5	17.13%	20.74%	0.826
15	12.75%	23.62%	0.540
30	19.33%	20.95%	0.923
60	18.62%	20.65%	0.902
100	23.23%	20.36%	1.141

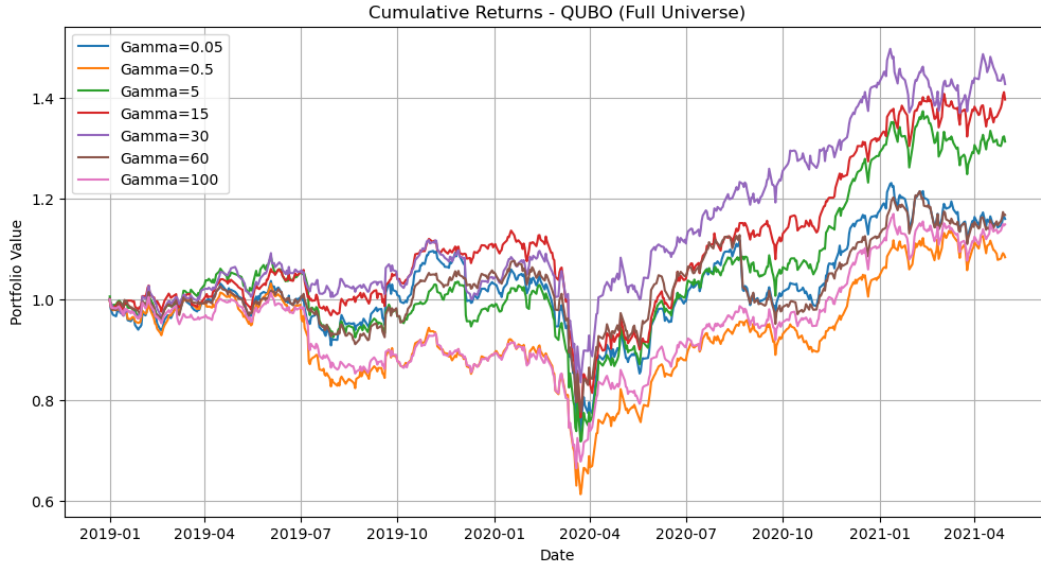


Figure 9: Out-of-sample cumulative portfolio value for the QUBO mean-variance portfolio (Full Universe) across all seven risk-aversion levels γ . The $\gamma = 30$ curve (red) achieves the highest terminal value and Sharpe ratio. Performance is non-monotonic in γ : the intermediate $\gamma = 0.5$ (orange) underperforms both lower and higher values due to a less favorable 10-asset combination being selected.

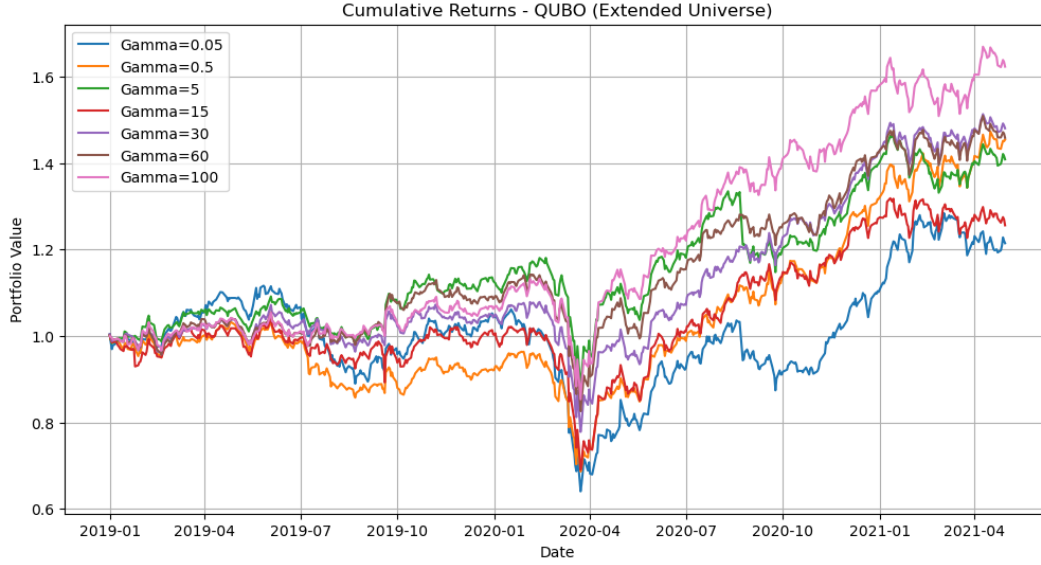


Figure 10: Out-of-sample cumulative portfolio value for the QUBO mean-variance portfolio (Extended Universe) across all seven risk-aversion levels γ . The $\gamma = 100$ curve (pink) achieves the highest terminal value and Sharpe ratio of 1.141, substantially outperforming the lower- γ configurations and the Markowitz benchmarks.

Out-of-sample Sharpe ratio of the QUBO mean-variance portfolio as a function of the risk-aversion parameter γ (log scale). Performance is non-monotonic in γ in both universes, with the Full Universe peaking at $\gamma = 30$ and the Extended Universe peaking at $\gamma = 100$.

Unlike the smooth efficient frontier traced by continuous-weight Markowitz, the QUBO's Sharpe ratio is *non-monotonic* in γ , because the asset selection is discrete and equally weighted, intermediate γ values can select a worse 10-asset combination than either a more return-focused or more risk-focused setting.

6.4 Stress Test: Discretized Portfolio Weights

As a scalability stress test, the binary selection model is extended to approximate *continuous* weights directly within the QUBO, by encoding each asset’s weight with B binary “bits”:

$$w_i = w_{i,\min} + (w_{i,\max} - w_{i,\min}) \sum_{k=1}^B c_k x_{i,k}, \quad c_k = \frac{1}{2^k}, \quad (12)$$

with $w_{i,\min} = 0$, $w_{i,\max} = 0.2$ (a 20% single-name cap), turning the N -asset problem into an $N \times B$ -variable QUBO with budget constraint $(\sum_i w_i - 1)^2$ replacing the cardinality penalty. At $B = 3$ this produced **11,325** QUBO terms for the Full Universe and **6,555** for the Extended Universe, an order of magnitude larger than the cardinality-only formulation (1275 / 741 terms), illustrating how quickly problem size grows once weights, not just selection, are binarized.

Table 10: Top selected weights, discretized QUBO ($B = 3$, initial parameters $\gamma=5$, $\lambda=50$)

Full Universe (top 5 of 10, equal $\approx 3.8\%$ each)			Extended Universe (top 5 of 10, equal $\approx 3.8\%$ each)		
BHARTIARTL,	CIPLA,	COALINDIA,	ASIANPAINT,	BAJFINANCE,	BHAR-
DRREDDY, GAIL			TIARTL, BPCL,	BRITANNIA	

With the initial parameter settings, the model produced nearly uniform portfolio weights (approximately 3.8% across 10 selected assets). This occurred because the low bit resolution ($B = 3$) and strong constraint penalties limited the model’s ability to assign different weights. After reducing γ and λ and increasing the scale of the predicted returns ($\gamma = 1$, $\lambda = 5$, returns scaled by $10\times$), the portfolio became more differentiated. For example, the top five Full-Universe stocks (BHARTIARTL, COALINDIA, HEROMOTOCO, MARUTI, and SUN-PHARMA) each received weights of about 8.05%, while the remaining stocks received about 3.45%. This demonstrates that the formulation can generate non-uniform allocations, although achieving meaningful weight differentiation requires careful tuning of the model parameters. Since this approach was primarily used to evaluate feasibility and scalability, and was not tested under the common out-of-sample backtesting framework, it is not included in the benchmark comparison in Section 7.

7 Benchmark: Comparative Analysis Across All Methods

7.1 Comparability

Four portfolio construction methods were evaluated using the same out-of-sample testing framework: **Equal Weight**, **Classic Markowitz**, **ML Markowitz** (Section 4), and the **QUBO Mean-Variance** portfolio (Section 6). All four use the same train/test split, rebalance date (2019-01-01), test-period returns, and performance metrics, allowing for a direct comparison. The results from Section 3 and the return-only QUBO model in Section 5 are based on different evaluation settings and are therefore discussed separately rather than included in the benchmark comparison.

7.2 Master Comparison: Best Out-of-Sample Sharpe Ratio per Method

For Classic and ML Markowitz, the best-Sharpe risk target from Tables 3–4 is shown; for the QUBO portfolio, the best-Sharpe γ from Tables 8–9.

Table 11: Best out-of-sample configuration per method, both universes

Method	Full Universe			Extended Universe		
	Ret. (%)	Vol. (%)	Sharpe	Ret. (%)	Vol. (%)	Sharpe
Equal Weight	15.06	24.33	0.62	14.39	25.22	0.57
Classic Markowitz	13.49	20.02	0.67	16.24	21.94	0.74
ML Markowitz	13.15	19.36	0.68	17.03	21.56	0.79
QUBO Mean-Variance	17.51	19.95	0.878	23.23	20.36	1.141

Figure 11 provides a direct visual comparison of the best out-of-sample Sharpe ratio achieved by each method across both universes.

The covariance-aware QUBO portfolio achieved the highest out-of-sample Sharpe ratio among the methods evaluated in both asset universes. It also exhibited a competitive risk-return profile, with realized returns exceeding those of the alternative approaches while maintaining similar levels of volatility.

7.3 Interpreting the Benchmark Results

Several factors may explain the stronger out-of-sample Sharpe ratio achieved by the QUBO mean-variance portfolio:

- **Simpler portfolio construction.** By selecting exactly 10 assets with equal weights, the QUBO approach avoids estimating individual portfolio weights. In contrast, the Markowitz models must determine both which assets to select and how much to allocate to each, making them more sensitive to errors in estimated returns and covariances.
- **Focused asset selection.** The concentrated 10-asset portfolio can benefit when the selected assets perform well during the test period. The predictive ability of the ML model suggests that the asset ranking contained useful information, which may have contributed to the portfolio’s performance.
- **Limited evaluation period.** The results are based on a single rebalance date and one out-of-sample test period. As such, the observed performance should be viewed as evidence from one backtest rather than a guarantee that the same ranking would hold across different market conditions.

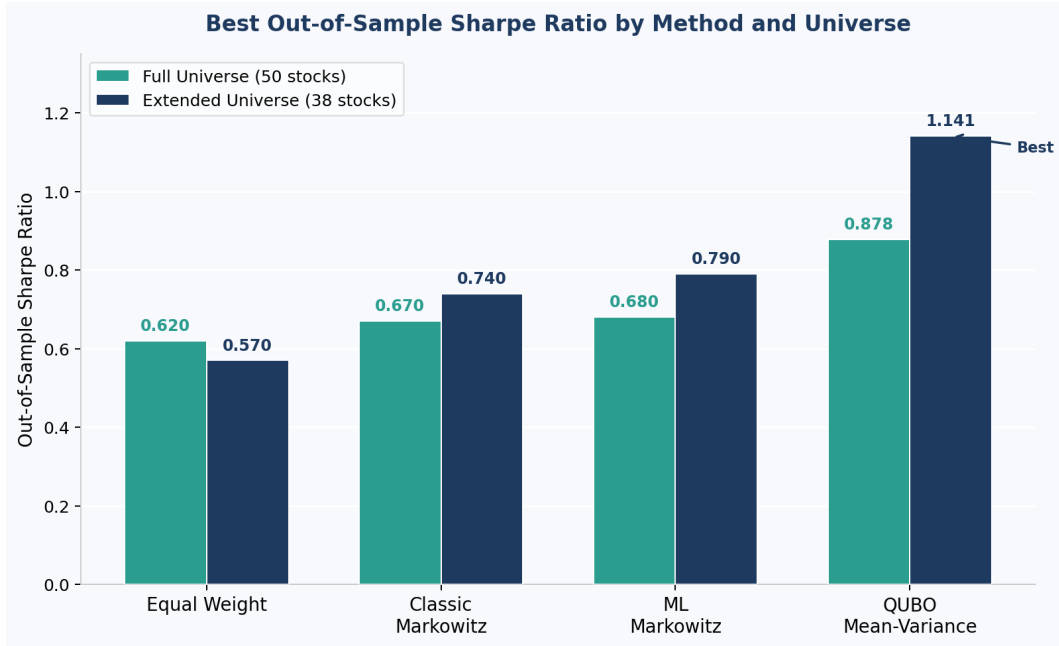


Figure 11: Best out-of-sample Sharpe ratio by method and asset universe. The QUBO Mean-Variance portfolio achieves the highest Sharpe ratio in both the Full Universe (0.878 at $\gamma = 30$) and the Extended Universe (1.141 at $\gamma = 100$), outperforming ML Markowitz, Classic Markowitz, and Equal Weight benchmarks. Note that each method is shown at its own optimal hyperparameter setting, so the comparison reflects best-case performance rather than a fixed-parameter evaluation.

Caveats. The comparison is not perfectly like-for-like: the QUBO methodology trades away weight optimization for a hard cardinality constraint, so it answers a different practical question (“best 10-name equal-weight portfolio”) than Markowitz (“best portfolio at this risk budget”). Simulated annealing is also a classical heuristic with no global-optimality guarantee at full scale (it is being used here as a proxy for the kind of search a quantum or quantum-inspired solver, e.g. QAOA [11], would perform); a different random seed or annealing schedule could shift results, and the non-monotonic γ -sensitivity indicates the result is not fully stable across parameter choices either. Both results nonetheless indicate that cardinality-constrained, risk-aware QUBO selection is a genuinely competitive, not merely a “discretized approximation of”, alternative to continuous-weight mean-variance optimization for this asset universe and period.

7.4 Computational Scalability

Brute-force QUBO solving is only feasible at small scale ($2^{15} = 32,768$ enumerable on the reduced problem; $2^{50} \approx 10^{15}$ infeasible at full scale). Simulated annealing comfortably handles the cardinality-only QUBO (1275 / 741 terms) and the mean-variance QUBO (similar size) at full scale. The discretized-weight stress test, however, shows how quickly the problem grows once weights are binarized: 11,325 terms (Full) / 6,555 terms (Extended) at just $B = 3$ bits per asset, a regime where the promised advantage of genuine quantum or quantum-inspired hardware over classical heuristics like simulated annealing would become most relevant.

8 Future Work

The analyses presented in this report suggest several concrete directions for extending and stress-testing the findings. These are listed roughly in order of priority, from the most immediate gap to the more ambitious long-term directions.

8.1 Rolling Backtests with Periodic Refitting

The most pressing limitation of the current study is that all out-of-sample results derive from a single portfolio construction date (January 1, 2019) and a single holding period. This means that the QUBO portfolio’s Sharpe ratio advantage over ML and Classic Markowitz could reflect a favorable draw from one particular market environment rather than a robust, repeatable edge.

The natural fix is a rolling backtest: rebalance the portfolio monthly, and at each rebalance date refit the ML model on all data available up to that point, re-estimate the covariance matrix on the same expanding or rolling window, and re-solve the QUBO. This approach is important for two reasons. First, without refitting, the ML model becomes progressively stale — its parameters were estimated on pre-2019 data and have no way to incorporate the structural shift caused by the COVID-19 shock of March 2020. Second, monthly rebalancing matches the 15-day prediction horizon used by the ML model, so the forecasts remain within their intended range at each decision point. A rolling backtest across 20–30 rebalance dates would give a distribution of realized Sharpe ratios for each method, making the comparison far more statistically meaningful than the single-window result reported here.

8.2 Transaction Costs

The current report evaluates all portfolios on gross returns, with no deduction for the costs incurred when changing holdings. This is a significant omission for the QUBO portfolio in particular. Because the QUBO selects exactly 10 assets with equal weights, each monthly rebalance may rotate several names in and out of the portfolio entirely, generating 100% turnover on those positions. In contrast, the continuous-weight Markowitz portfolios tend to make smaller marginal adjustments to existing holdings.

Once rolling backtests are in place, the straightforward next step is to apply a realistic transaction cost model, a fixed cost per trade plus a market-impact term proportional to trade size, and recompute net-of-cost Sharpe ratios. It is plausible that transaction costs narrow or eliminate the QUBO’s gross-return advantage, particularly at higher rebalancing frequencies. Incorporating a turnover penalty directly into the QUBO objective (as an additional quadratic term discouraging changes from the previous period’s selection) is one way to address this within the existing framework.

8.3 Alternative and More Expressive ML Models

The return forecasting model used in this report is a pooled cross-sectional linear regression, chosen for its simplicity and transparency. Its out-of-sample MSE of 0.011–0.012 reflects moderate predictive accuracy, and the cross-sectional ranking it produces is informative enough to be useful, but there is clearly room to improve the signal.

Two directions are worth pursuing. The first is gradient boosted trees (e.g. XGBoost [12] or LightGBM [13]), which can capture non-linear interactions between features, for instance, the interaction between momentum and volatility regime, that a linear model cannot. Crucially, these models remain interpretable via tools such as SHAP (SHapley Additive exPlanations) [5], which decompose each prediction into the contribution of individual features. This interpretability is important in a portfolio context: a model that ranks BHARTIARTL above HINDUNILVR should be able to say *why*, in terms of observable features, rather than producing a black-box

score. The second direction is a richer feature set: sector membership, valuation ratios, earnings surprise, and macroeconomic indicators could all add cross-sectional information beyond the price-based features currently used. A more expressive model with more features increases the risk of overfitting, so any extension should be evaluated with the same strict train/test discipline applied here.

8.4 QAOA and D-Wave Quantum Annealing

The simulated annealing solver used throughout this report is a classical heuristic that approximates the behavior of a quantum optimizer. It was chosen because it is accessible, reproducible, and scales to the problem sizes considered here. However, the report is explicitly motivated by quantum-inspired optimization, and a natural next step is to run the same QUBO formulations on actual quantum hardware.

Two platforms are relevant. D-Wave’s quantum annealers [10] operate on the same physical principle as simulated annealing, minimizing an energy function encoded as a QUBO, but exploit quantum tunneling to escape local minima more efficiently than classical thermal fluctuations allow. The QUBO matrices constructed in Sections 5 and 6 could be submitted to D-Wave’s Leap cloud platform with minimal reformulation, making this the most direct extension of the existing work. QAOA (Quantum Approximate Optimization Algorithm) [11], by contrast, runs on gate-based quantum computers such as those provided by IBM Quantum, and represents a more flexible but also more technically demanding approach. A controlled comparison, same QUBO, solved by simulated annealing, D-Wave, and QAOA — would directly quantify whether real quantum hardware provides a solution-quality or speed advantage for portfolio-scale problems at current qubit counts and noise levels.

8.5 Multi-Period Optimization

All methods in this report solve a single-period problem: given today’s estimates, what is the best portfolio to hold from now until the next rebalance? In practice, an investor also cares about the *sequence* of portfolios over time, specifically, about avoiding costly reversals where a stock is bought this period and sold next period simply because the optimizer changed its mind slightly.

Multi-period optimization addresses this by solving for a sequence of portfolio decisions jointly, penalizing transitions that generate high turnover. In the Markowitz setting, this leads to a dynamic programming or model-predictive control formulation. In the QUBO setting, it means adding inter-period terms to the objective that link successive binary selection vectors, discouraging large changes in the selected set. This is considerably more complex to implement and solve than the single-period problems studied here, but it would bring the framework closer to the requirements of a real deployment, where transaction costs, tax considerations, and tracking error constraints all create incentives to limit portfolio turnover.

9 Conclusion

Across the study, the portfolio optimization problem was studied from several perspectives. The work began with the classical Markowitz mean-variance framework (Section 2), followed by convex optimization methods for practical portfolio construction (Section 3). Machine learning techniques were then used to improve return estimation (Section 4), and finally QUBO-based formulations were explored for discrete portfolio selection, first without risk and then with covariance information incorporated (Sections 5 and 6).

The benchmark results show that, for the dataset and test period considered, the covariance-aware QUBO portfolio achieved the highest out-of-sample Sharpe ratio among the methods evaluated in both asset universes. One possible reason is that restricting the portfolio to a small number of equally weighted assets reduces sensitivity to errors in estimated returns and covariances. However, this comes at the cost of reduced flexibility compared with continuous-weight Markowitz optimization, which allows more precise control over portfolio allocations and risk. Overall, the results suggest that QUBO-based approaches can be effective when discrete constraints such as a fixed number of holdings are important, while classical optimization remains a strong and widely applicable approach for continuous portfolio construction.

Key limitations to keep in mind: All out-of-sample results in this report come from a single rebalance date and a single backtest window; simulated annealing is a classical heuristic without optimality guarantees at scale; equal weighting was used as a simplifying convention within the QUBO-selected subset rather than an optimized choice; and the ML return predictions, while modestly informative at a 15-day horizon, remain noisy. A natural next step would be to backtest all four methods across multiple rolling rebalance dates to assess whether the QUBO portfolio’s advantage persists out of a single-window result.

Code Repository

The complete implementation, datasets, experiment notebooks, and report source code used in this project are available at:

<https://github.com/shyam-lab/portfolio-optimization-nifty50/>

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Glossary

Cardinality constraint

A restriction on the number of assets held in a portfolio; here, the QUBO formulations enforce exactly $K = 10$ selected assets.

Covariance matrix (Σ)

A matrix describing the variances of individual asset returns and the pairwise co-movement between assets.

Efficient frontier

The set of portfolios offering the highest expected return for a given level of risk, or equivalently the lowest risk for a given expected return.

Markowitz optimization

Mean–variance portfolio optimization that balances expected return against portfolio variance [1].

QUBO

Quadratic Unconstrained Binary Optimization, an objective of the form $x^\top Qx$ with binary decision variables $x_i \in \{0, 1\}$ [8,9].

Risk-aversion parameter (γ)

The coefficient controlling the strength of the variance penalty in the risk-adjusted objective.

Sharpe ratio

A risk-adjusted performance measure, reported here as annualized portfolio return divided by annualized volatility.

Simulated annealing

A stochastic classical optimization heuristic used here to approximately solve full-scale QUBO instances [6,7].